BAD BLOOD V2

Blood Pressure Models, Training, and Personalization

From many people's recordings to one person's model

Jingyi Guo
Bad Blood Project, University of Notre Dame

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What the model predicts

Blood pressure (BP) has two familiar numbers, such as 120/80.

- **Systolic blood pressure (SBP)** is the higher pressure during a heartbeat.
- **Diastolic blood pressure (DBP)** is the lower pressure between beats.
- **mmHg** means millimeters of mercury, the pressure unit.

Our model reads a **10-second segment**: ten seconds of signals from **one person**.

The output is two numbers for that segment: estimated SBP and DBP.

[7] The released models estimate segment-level pressure. A segment is also called a window.

PulseDB contains three waveforms and pressure labels

Whose data? Hospital participants in the public dataset. Each segment belongs to one person.

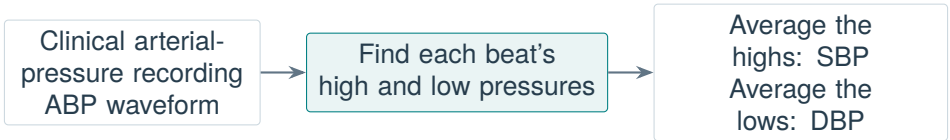
Field	Full name	What it records
ECG	Electrocardiogram	Heart electrical activity over time
PPG	Photoplethysmogram	Optical changes related to blood volume over time
ABP	Arterial blood pressure	Pressure inside an artery over time
SBP, DBP	Systolic, diastolic blood pressure	Two scalar labels summarizing each segment

A waveform is a sequence of samples. A scalar label is one number.

[1] PulseDB includes recorded waveforms as well as derived labels. Its files also contain participant identifiers and other metadata.

Where ABP comes from, and how it gives the labels

Whose data? The same hospital participant and recording interval as the ECG and PPG.

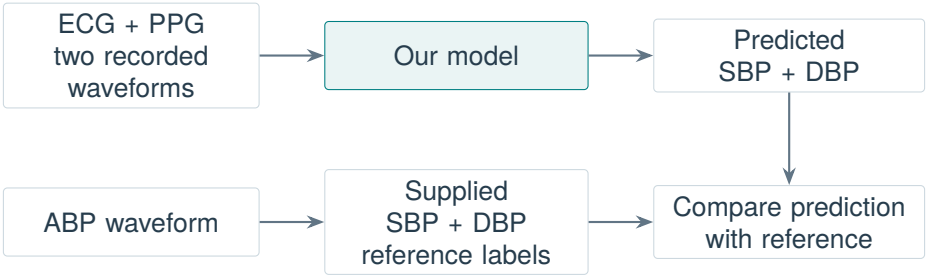


- PulseDB carries the ABP waveform from its clinical source recordings.
- It derives each 10-second segment's labels from the beats in that segment.
- Our training code reads those supplied SBP/DBP labels.

ABP is recorded source data. We do not reconstruct it by interpolating SBP/DBP.

[1] PulseDB, Sections 2.2, 2.4.3 and 2.5. The two released models predict scalar labels rather than an ABP waveform.

What PulseDB provides versus what our model uses



The **PPG-only version** uses only the optical waveform. Neither version needs current ABP or current true BP as a prediction input.

During training, the comparison updates weights. During testing, it only scores the saved model.

[1] Label provenance. [7] Actual model inputs and scalar regression target. Later slides explain the extra earlier reference measurements used for personalization.

Public hospital data and our future student measurements

	Current public-data experiments	Planned Notre Dame measurements
People	Hospital participants	Student volunteers
Model signals	ECG and PPG recordings	ECG and PPG from our sensors
Reference pressure	Labels from recorded ABP	Intermittent upper-arm cuff readings
Continuous ABP	Present in PulseDB	Not collected by our proposed equipment
Results in this deck	All reported results	No student results reported here

The current personal models also use public-data participants.

[1,7] Current experiment provenance. A student session with a few cuff readings supplies fewer reference labels than the dense public-data calibration studied here.

Whose data enters each of the four stages?

Stage	Whose recordings?	What happens to the model?
1. Group training	Many public-data participants	Learn a shared waveform-to-pressure model
2. Group pair training	Same training group. Each pair comes from one person	Learn how pressure changes between two signal states
3. Personal fitting	Only person P's early recordings and known pressure labels	Copy the shared model, then fit its last CNN block, SE and change head to P
4. Personal testing	P's later signals, plus P's saved early references	Freeze weights. Predict P's later pressure and score it

Many people teach the starting model. One person's early data adapts their own copy.

[7] Stages 1 and 2 share source-training participants. Each target person gets an independent personal fit.

The people and the jobs of their data

Whose data? All groups come from the VitalDB-derived portion of PulseDB, not ND students.

Group	People	Windows	Job
Group training	1,085	390,600	Learn shared weights
Group validation	192	69,120	Choose population checkpoints
Group main test	141	56,400	Score uncalibrated predictions
BP-range stress test	116	666	Different pressure range
Personal development	30	8,882	Each person's own early 80%: train and select settings
Personal test	Same 30	2,211	Each person's later 20%: score only

Every target person selects settings using their own development data.

[7] A 10-second temporal guard removes boundary windows. Same 30 people across modes and seeds. These target records were previously examined; the current comparison is exploratory.

How the data is stored

- **PulseDB** reorganizes recordings from VitalDB and MIMIC-III, another clinical database. We currently use its **VitalDB-derived portion**.
- Original **MATLAB .mat files** contain waveforms, labels and participant information. Individual-record files also provide sample times in `T`.
- Our prepared `waves.npy` is a NumPy array with shape $N \times 2 \times 1250$.
- Our `y.npy` has shape $N \times 2$, holding SBP and DBP labels in the same window order.

N = number of windows. Each window contains **two signal channels** and **1,250 samples per channel**.

The training collection has many people. An individual row is still one person's window.

[1] Official file documentation. [7] shared/data.py. NumPy is the numerical-array library used by our Python pipeline. MIMIC-III: Medical Information Mart for Intensive Care III.

Model words used throughout this presentation

Term	Meaning here
CNN	Convolutional neural network. Learned filters scan the waveform.
1D	One-dimensional. The filters move along the time axis.
SE	Squeeze-and-Excitation. A small module weights learned feature traces.
Encoder / backbone	The shared part that converts a waveform into 48 summary numbers.
Head	The final part that converts features into a prediction.
MLP	Multilayer perceptron. A stack of fully connected layers.
PairSE	Our project's name for a paired-reference model built around CNN+SE.

[2,3,7] PairSE is our implementation name. It is not the title of the published method that inspired it.

Training words used throughout this presentation

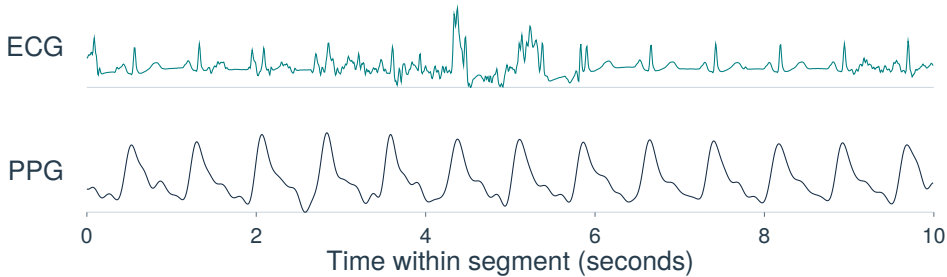
Term	Meaning here
Feature	A number computed from a window. It changes with the input.
Parameter / weight	A model value learned during training and saved afterward.
Batch	A set of windows used together for one weight update.
Step	One optimizer update of model weights.
Epoch	One pass through the group training windows.
Checkpoint	A saved version of model weights after training.
Seed	A number controlling random initialization and sampling.
Pretraining	Learning the shared starting model from a group.
Personal fitting	Updating a copy using one person's early labeled data.

Running the same model three times uses three seeds, not three new datasets.

[7] Our experiments use seeds 42, 43 and 44. All three use the same participant split.

One real person's ten seconds of input

Public test participant p000053_1, row 0



Each channel has **1,250 samples at 125 hertz (Hz)**: 125 measurements per second. The vertical axes show normalized signal amplitude.

Reference label for the whole segment: **115.12 / 50.08 mmHg**.

Real recorded data, first eligible test row chosen by file order. The plot shows all 1,250 samples per channel. One example explains mechanics, not aggregate model performance.

Preparing one window as a numerical array

Use the supplied filtered ECG and PPG.
Scale each channel inside each window:

$$x'_t = \frac{x_t - \min(x)}{\max(x) - \min(x)}.$$

- Reject nonfinite or flat inputs.
- Preserve sample order and channel order.
- Stack ECG first, then PPG.
- Store numbers in 32-bit floating-point format (`float32`).

First six values of the real window

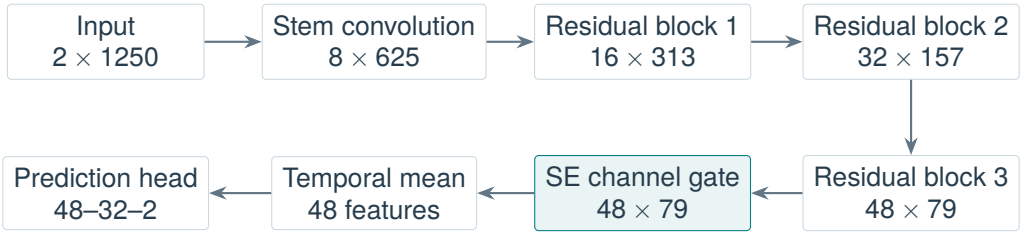
Index	Time (s)	ECG	PPG
0	0.000	0.36577	0.29328
1	0.008	0.37430	0.28963
2	0.016	0.39656	0.28925
3	0.024	0.41135	0.29156
4	0.032	0.40577	0.29585
5	0.040	0.41031	0.30114

One window: 2×1250.
A batch of 256: $256 \times 2 \times 1250$.
The batch can contain windows from different training people.

[7] shared/data.py. PulseDB supplies filtered normalized fields. We do not derive model inputs from ABP or fit normalization using later BP labels.

The group model processes one window at a time

The same learned weights serve every person. Each box below processes **one person's current window**.



Shapes show **learned feature traces × time positions**. The initial 2 means ECG and PPG. The later 48 means learned traces, not 48 sensors or people.

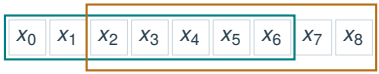
38,296 trainable parameters. Each window produces one SBP and one DBP.

[7] implementation/shared/models.py. The input retains all 1,250 samples before the learned stride-2 stem.

Convolution: a small filter scans the signal

A small learned filter slides across time. The first layer uses **7 samples per channel** at each position.

$$u_{k,t} = \sum_{c=1}^{2} \sum_{j=0}^{6} w_{k,c,j}\, x_{c,2t+j-3}.$$



The same filter weights serve every time position.

A filter's weights are learned across many training participants. The resulting feature values depend on this particular input window.

- 8 filters create 8 feature traces.
- Stride 2 moves two samples at a time.
- Padding 3 supplies zeros at the edges.

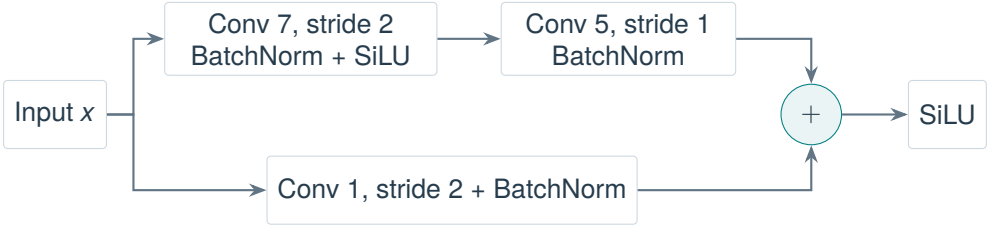
[7] Actual stem: Conv1d(2,8,kernel=7,stride=2,padding=3,bias=False), BatchNorm, SiLU. The small diagram illustrates indexing, not measured signal values.

The small operations inside the network

Diagram label	Meaning
Conv	Convolution. A learned filter scans nearby time samples.
BatchNorm	Batch normalization. It scales intermediate features using training-batch statistics and learns an adjustment.
SiLU	Sigmoid Linear Unit, $u/(1+e^{-u})$. A smooth nonlinear transformation.
ReLU	Rectified Linear Unit, $\max(0,u)$. It keeps positive values and sets negative values to zero.
Sigmoid	A function that maps a number into the interval 0 to 1.
Linear / dense	A fully connected layer. It mixes all input features using learned weights.

[7] Operations in shared/models.py and paired_se/model.py. At prediction time, batch normalization uses saved training statistics.

Inside one residual block

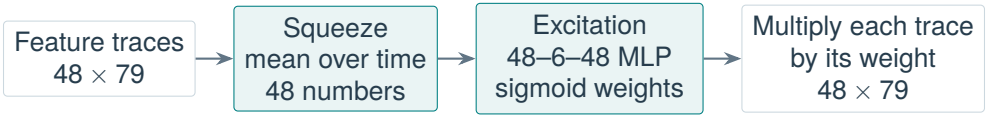


$$x_{\text{next}} = \mathrm{SiLU}\big(F(x) + P(x)\big).$$

- $F(x)$ learns a transformation. $P(x)$ carries a matching shortcut.
- The projection P adjusts channel count and time length before addition.
- Our three blocks increase channels: $8 \to 16 \to 32 \to 48$.

[2] He et al., residual learning. [7] Our 1D kernels, projection shortcuts and SiLU implementation.

Squeeze-and-Excitation: weighting learned traces

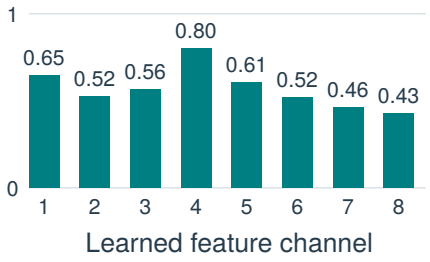


$$z_c = \frac{1}{79}\sum_t U_{c,t}, \qquad a = \mathrm{sigmoid}(W_2\,\mathrm{ReLU}(W_1 z + b_1) + b_2)\,, \qquad \widetilde{U}_{c,t} = a_c U_{c,t}.$$

- **Squeeze**: summarize each learned trace over the window.
- **Excitation**: a small multilayer perceptron sets 48 weights from the summary.
- Apply one weight to all 79 positions of the corresponding trace.

[3] Hu et al., CVPR 2018. Our 1D adaptation uses temporal pooling and a reduction ratio of 8. Added parameters: 630, about 1.7% of the plain CNN.

One person's window produces these attention weights



First 8 of 48 gates

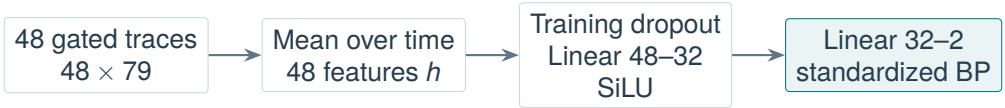
The model learned its parameters from the training group. These gate values come from test person `p000053_1`'s example window.

- Channel 4 gets a multiplier of about 0.80.
- Channel 8 gets a multiplier of about 0.43.
- A different input window can produce different weights.

These are feature-channel weights, not ECG-versus-PPG importance scores.

[7] Saved dual-input seed-42 model. A gate is a computed value, not a newly fitted personal parameter. Channel meanings emerge during training.

Converting 48 features into two pressure estimates



$$\widehat{y}_{\mathrm{mmHg}} = \mu + \sigma \odot \widehat{y}_{\mathrm{standardized}}.$$

Real example	SBP	DBP
Network output	0.5066	0.5820
Training-label mean μ	115.41765	62.92810
Training-label standard deviation σ	18.94188	12.08322
Prediction (mmHg)	125.01	69.96
Reference (mmHg)	115.12	50.08

[7] Mean and standard deviation come from the group training labels. Dropout randomly sets 15% of features to zero during training and is off during prediction. $\odot$ means multiply matching entries.

Training loss: how a wrong answer changes the model

Mean squared error (MSE) averages squared prediction errors. Standardization first subtracts the training-group mean and divides by its standard deviation.

$$z_{b,d} = \frac{y_{b,d} - \mu_d}{\sigma_d}, \qquad L = \frac{1}{2B} \sum_{b=1}^{B} \sum_{d \in \{\mathrm{SBP}, \mathrm{DBP}\}} (\widehat{z}_{b,d} - z_{b,d})^2.$$

Why standardize the labels?

- B is the batch size. b indexes a window and d a pressure output.
- Standardization sets their scales before the joint loss.
- Squaring gives larger errors more influence on an update.

One saved-model example

True z: -0.0157 / -1.0634

Predicted z: 0.5066 / 0.5820

Two-output MSE: **1.4901**

[7] MSE is the training objective. Later slides define the separate evaluation measures. This calculation uses one test window to illustrate the arithmetic, without updating the saved model.

Stage 1: one update using the training group

Whose data? A shuffled batch from the 1,085 training people, not one target person's calibration bank.

1. Take **256 training windows** and their supplied SBP/DBP labels.
2. Add input noise with standard deviation **0.003**.
3. Run the CNN, SE gate and readout to obtain predicted SBP/DBP.
4. Compute standardized MSE against the observed labels.
5. Backpropagation calculates how each weight affects the loss.
6. The optimizer updates the network weights using those gradients.

[7] An optimizer is the numerical rule for changing weights. Backpropagation computes gradients: how a small weight change would affect the loss.

The optimizer and training settings

Setting	Role and actual value
AdamW optimizer	Adaptive Moment Estimation with decoupled weight decay. It scales updates using past gradients.
Learning rate	Overall update size: **0.001**.
Weight decay	Shrinks weights during group training: **0.001**.
Gradient clipping	Caps the total gradient norm at **5** to limit a large update.
Batch size	**256 windows** per step.
Training loss	Mean squared error of the two standardized BP labels.

The validation group selects checkpoints. Its labels do not contribute optimizer updates.

[7] attention_cnn/train.py. The personal optimizer later uses Adam without weight decay.

How long group training actually ran

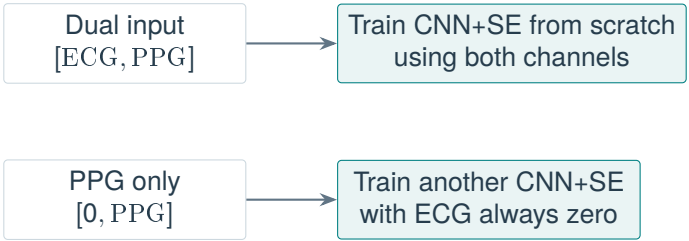
- Maximum: **40 epochs**.
- Stop after **8 epochs** without validation improvement.
- Halve learning rate after **3** validation non-improvements.
- Score: mean absolute error (**MAE**), averaged first per person, then across people and the two pressures.

Input	Seed	Run	Best	Val. MAE
dual	42	9	1	10.291
dual	43	9	1	10.457
dual	44	9	1	10.421
ppg	42	9	1	10.374
ppg	43	9	1	10.583
ppg	44	9	1	10.406

All six fits ran **9 epochs** and retained **epoch 1**. Additional epochs did not improve this validation criterion.

[7] Recorded completions. Run and Best are epoch numbers. Three seeds describe fitting variation, not three independent participant cohorts.

Two separately trained group models



- Both have **38,296 parameters**, identical settings and seeds 42/43/44.
- The PPG-only model learns without ECG throughout training.
- Its internal input mask also applies inside the personal reference encoder.

[7] implementation/modality_se/model.py. Keeping two tensor slots makes the architecture comparison parameter-matched. The PPG inference wrapper can accept a single channel.

Why this group architecture?

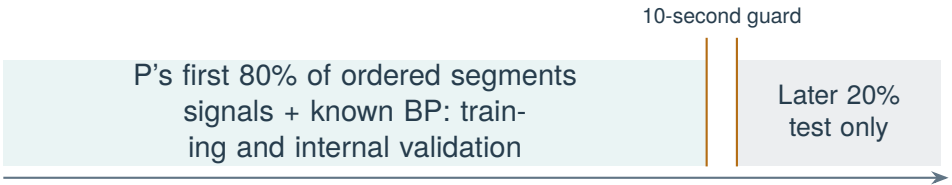
Source	Published idea	Our adaptation
Our V1 baseline	Learn from complete signal windows	Keep all 1,250 input samples before learned downsampling
He et al. [2]	Residual learning: a shortcut around a transformation	Three compact one-dimensional blocks, widths 16, 32, 48
Hu et al. [3]	Squeeze-and-Excitation: weight feature channels	One gate after the final block, with a 48–6–48 network

We combine published building blocks with our own compact signal model.

[2,3,7] These are adaptations of ideas, not reproductions of the original image-classification architectures. The six-architecture comparison appears in the results section.

One person's timeline

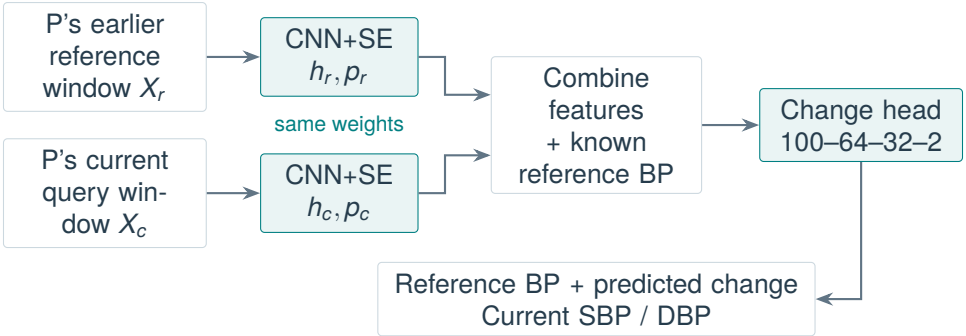
Whose data? Only person P. All 30 personal pipelines repeat this procedure independently.



- The **reference bank** stores P's development waveforms and known SBP/DBP.
- Three time-ordered validation periods inside the 80% choose P's settings.
- After selection, refit on all of P's development bank, then freeze the model.

[7] 20261015 protocol: chronological segment-count split, not a random split or the previous 20%-of-elapsed-time support rule. Later true BP is available only to the scoring code.

PairSE compares two windows from the same person



h contains **48 learned features**. *p* contains **two group-model pressure estimates**. Subscript *r* means reference and *c* means current.

The reference tells the model this person's known BP at a known signal state.

[4,5] Related paired-measurement and reference-change ideas. [7] Our shared CNN+SE and PairSE implementation. Both branches reuse one encoder, not two independently trained networks.

What goes into the 100-dimensional change head?

One reference and one current window from the same person.

Component	Size	Information supplied
$h_c - h_r$	48	How the current waveform features differ from the reference
h_r	48	The reference signal's feature state
y_r	2	Known reference SBP and DBP, standardized
$p_c - p_r$	2	The original model's expected BP change
Concatenated input	**100**	$48 + 48 + 2 + 2$

$$g_\phi : \quad 100 \xrightarrow{\text{Linear+SiLU}} 64 \xrightarrow{\text{Linear+SiLU}} 32 \xrightarrow{\text{Linear}} 2.$$

We join the four groups of numbers in order. Feature scaling uses statistics learned from the training group. A 0.01 minimum scale avoids dividing by a very small number.

8,610 change-head parameters; 46,906 in full PairSE.

[7] implementation/paired_se/model.py. Feature vectors in this slide denote standardized encoder features. BP uses its own source-label mean and SD.

The prediction equation, one term at a time

All terms describe **the same person**. y_r is their measured earlier BP. p_r, p_c are the shared model's earlier/current predictions. g_ϕ is the small change network with weights ϕ.

$$\widehat{y}_{c|r} = \underbrace{y_r}_{\text{known BP}} + \underbrace{(p_c - p_r)}_{\text{base-model change}} + \underbrace{g_\phi(z_{cr}) - g_\phi(z_{rr})}_{\text{learned change correction}}$$

$$z_{cr} = [h_c - h_r, h_r, y_r, p_c - p_r], \qquad z_{rr} = [0, h_r, y_r, 0].$$

- $g_\phi(z_{cr})$ learns a correction from the current/reference pair.
- $g_\phi(z_{rr})$ uses the reference as both inputs: an identity anchor.
- If $X_c = X_r$, both change terms cancel, giving exactly $\widehat{y}_{c|r} = y_r$.

Predict a change relative to a known measurement, rather than relearning absolute BP from scratch.

[7] This equation and identity subtraction are our implemented design. BP terms use source-label standardization; encoder features use their own source statistics.

How the design includes simple bias correction

Before group pair training, we set the change network's last layer to zero. Then $g_\phi = 0$ and one same-person pair predicts

$$\widehat{y}_{c|r} = y_r + p_c - p_r.$$

At prediction time, average over this person's K early references:

$$\frac{1}{K}\sum_r \left[y_r + p_c - p_r\right] = p_c + \frac{1}{K}\sum_r (y_r - p_r).$$

- K is the number of reference windows from this person.
- The right-hand side is **simple personal bias correction**: add this person's average earlier prediction error to the current estimate.
- Training can then learn a **waveform-dependent** adjustment to that starting point.

PairSE adds a waveform-dependent adjustment to ordinary bias correction.

[7] The equality explains source-head initialization. It does not guarantee that source training or personal fitting will outperform the bias baseline.

Our adaptation of the paired-measurement idea

Mekonnen et al. call their method **Mixed Deduction Learning (MDL)**. It is the paper that directly inspired our reference/current pairing [4].

Design aspect	Mekonnen et al. [4]	Our PairSE [7]
Paired inputs	Earlier and later PPG measurements	Reference/current ECG+PPG or PPG-only windows
Meaning of two channels	Two PPG recordings in a measurement pair	ECG and PPG are sensor channels inside each window
Training strategy	Mix personal calibration data with other subjects' data	Pretrain encoder, train source pair head, then adapt the personal tail and head
Network design	Their CNN-based mixed deduction model	Shared residual CNN+SE plus a 100-D change MLP
Prediction construction	Learn from paired measurements	Explicit reference BP, identity subtraction, all-reference averaging

[4,7] Direct inspiration, with our own architecture and evaluation. Related work: Schlesinger et al. [5] study reference-based BP changes, and Leitner et al. [6] study transfer to individuals.

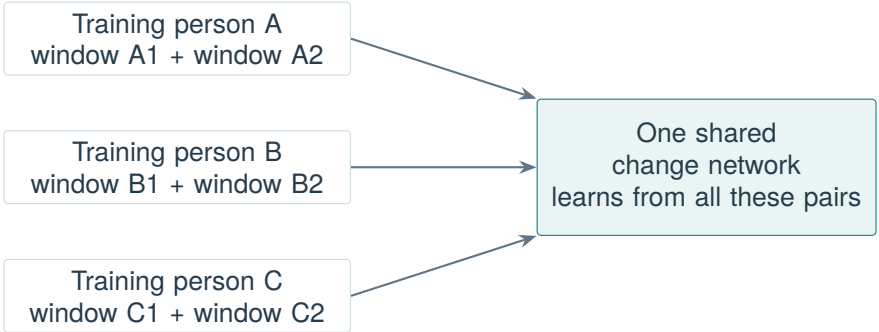
Stage 2 still uses the group, not a target person

Whose data? The same 1,085 training people. Every pair stays within one of those people.

1. Start from the saved population CNN+SE and **freeze its weights**.
2. Save 48 features and two pressure predictions for each training window.
3. For each training window, randomly choose a **different window from the same person** as its reference.
4. Give PairSE the reference's BP. Use the current window's BP as the loss target.
5. Train only the change head: Adam optimizer, learning rate **0.001**, batch **2,048**, standardized mean squared error, gradient clip **5**.
6. Choose a checkpoint using fixed validation pairs, up to **20 epochs** with patience **5**.

[7] paired_se/run.py. Each epoch visits 390,600 training windows. These group subset files lack timestamps, so pair order here is not chronological. Adam means Adaptive Moment Estimation.

Same-person pairs can still train a group model



Within a pair: one person. Across training: many people.

[7] Illustrative participants A/B/C show the implemented pairing rule. A pair never combines A's known pressure with B's current waveform.

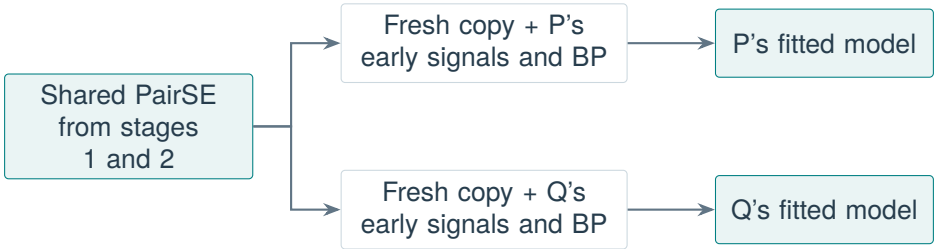
Group pair training: actual completed epochs

Input mode	Seed	Epochs run	Best epoch
ECG+PPG	42	20	18
ECG+PPG	43	17	12
ECG+PPG	44	16	11
PPG-only	42	20	16
PPG-only	43	20	15
PPG-only	44	20	20

The population backbone and the pair head have separate training schedules. These epochs use the group dataset. The next stage chooses the update count separately for each target person and pressure, using only their early data.

[7] Saved source-head completion records in paired_se/population and modality_se/pair_population. Each checkpoint is selected on source validation pairs.

Stage 3: one independent model copy per person



- P's updates affect only P's copy. Q starts from the same group checkpoint.
- Each person keeps their own early reference bank for later predictions.

[7] This is the target-person fitting procedure. It does not sequentially continue P's personalized weights on Q.

Exactly what changes during personal fitting

Whose data? Only this target person's development pairs; the starting weights come from the group.

Component	During this person's fitting
Stem + residual blocks 1 and 2	Frozen weights
Last residual CNN block	**Trainable** signal features
Last SE channel-attention gate	**Trainable** feature-channel weighting
Reference-conditioned change head	**Trainable** 100–64–32–2 network
Population readout weights	Frozen, but its input features can change
Label/feature scaling, BatchNorm statistics	Frozen; dropout disabled

33,336 trainable parameters per endpoint copy; 46,906 in full PairSE.

[7] Trainable names: change.*, encoder.body.6.*, encoder.gates.6.*. BatchNorm affine scale/bias in the last block can train. SE = Squeeze-and-Excitation; CNN = convolutional neural network.

Each person chooses their own fitting settings

Whose data? Only P's first 80%. No other target person chooses P's hyperparameters.

Internal period	Train on development bank	Validate on development bank
1	First 50%	Next 15% (50–65%)
2	First 65%	Next 15% (65–80%)
3	First 80%	Last 20% (80–100%)

These percentages are inside P's development bank. P's final test 20% is outside this entire procedure.

- Tail learning rate: **0.0001 or 0.0003**. Updates: **10,25,50,100**.
- Also compare zero extra updates; every candidate starts from the source model.
- Select SBP and DBP settings separately by mean MAE over the three periods.

[7] Ten-second guards at validation boundaries. Ties favor fewer updates, then lower learning rate. Hyperparameters are settings, not fitted weights. MAE = mean absolute error.

From selected settings to a fitted personal model

1. Save P's selected learning rate and number of updates for **SBP**.
2. Save P's separately selected settings for **DBP**.
3. Restart two copies of the same source model. Fit each endpoint on **all of P's development bank** using its chosen settings.
4. Each Adam step uses **every valid chronological reference/current pair**; reference end + 10 seconds must precede current start.
5. Optimize that endpoint's standardized MSE; clip gradient norm at 5; no weight decay.

Two endpoint checkpoints per person and source seed, using one tail-adaptation policy.

MSE = mean squared error. Adam = Adaptive Moment Estimation. The candidate with zero updates remains possible. The reference bank stays separate from the weights.

What is available at a later prediction time?

Whose data? The same person P, now at a later time. Fitting has finished.

Available to the predictor	Role
P's current ECG/PPG	The signal whose corresponding pressure we estimate
P's saved early signals	Reference waveform states
P's saved early SBP/DBP	Known pressures at those earlier states
P's fitted model weights	Convert each reference/current pair into an estimate

We average estimates over all eligible early references. The evaluation code then compares the output with P's **later true BP**, which the predictor never receives.

We observe the later ECG/PPG. The earlier 80% calibrates the model; the later signals are observed inputs.

[7] Saved personal weights and the separate personal reference bank jointly define the deployed personal predictor.

A real tail-adapted prediction, step by step

Person p000006, dual-input seed 42. First reference row 0; first eligible test row 97. This person's two endpoints each selected learning rate 0.0003 and 25 updates.

Quantity	SBP / DBP (mmHg)
Known reference BP y_r	108.29 / 66.40
Readout change $p_c - p_r$ after tail adaptation	-2.33 / -4.55
Learned change correction, converted to mmHg	-20.32 / -14.14
Prediction using this one reference	85.64 / 47.71
Actual inference: average over all 96 early references	**92.44 / 47.95**
True BP for the later query, used only for scoring	93.10 / 50.36

This early bank supplies **4,488 valid pairs** per fitting step. Pairing reuses observed labels; it does not create new independent measurements.

[7] Replayed 20261015 tail checkpoints. Example chosen by participant/row order, not by error. Each endpoint uses its own selected copy; arithmetic is converted from standardized BP to mmHg.

The two simple controls use exactly the same data

Personal mean: one constant prediction for this person

$$\widehat{y}_c^{\text{mean}} = \frac{1}{K} \sum_r y_r.$$

Bias correction: retain the population prediction's changes

$$\widehat{y}_c^{\text{bias}} = p(X_c) + \frac{1}{K} \sum_r \left[y_r - p(X_r) \right].$$

Same real query as the previous slide	SBP / DBP (mmHg)
Personal mean	94.97 / 49.00
Population prediction + personal bias	95.34 / 46.18
Tail-adapted PairSE with the same reference bank	92.44 / 47.95

Every method gets the same early labels, later queries and input modality.

[7] Personal mean ignores query waveforms. Bias correction uses the modality's own pretrained model. Aggregate results, rather than this one example, decide the comparison.

How we measure prediction error

Metric	Question answered	Better
MAE: mean absolute error	Average absolute error in mmHg	Lower
RMSE: root mean squared error	Square errors, average them, then take the square root	Lower
Signed Bias	Systematic overprediction or underprediction	Closer to 0
Error SD: standard deviation	Variation of errors around their mean	Lower
Within $\pm 5/10/15$	Percentage of estimates close to the reference	Higher
Participant-macro MAE	Average each person's MAE, then weight people equally	Lower

Macro versus pooled: macro weights participants equally. Pooled weights every window equally. Macro SD averages within-person SD.

We report the mean of **three separate seeds**, not predictions from a three-model ensemble.

[7] shared/metrics.py. SD uses the sample denominator n-1. These metrics characterize our dataset errors and do not establish clinical-device certification.

Group models: six matched architectures

Validation: **192 people**. Main test: **141 people**. Each saved group model predicts all their windows without personal fitting.

Architecture	Validation MAE	Main-test MAE
Plain residual CNN	**12.41** / 8.33	12.59 / 8.36
CNN + SE	12.51 / **8.26**	**12.39** / **8.00**
CNN + temporal gate	12.49 / 8.32	12.58 / 8.31
CNN + CBAM	12.52 / 8.29	12.42 / 8.11
CNN + additive gate	12.44 / 8.34	12.66 / 8.18
CNN + SE at all stages	12.47 / 8.32	12.49 / 8.23

SBP / DBP, mmHg. Bold: lowest MAE for each endpoint within each partition.

Observed test comparison: CNN+SE improves MAE from 12.59/8.36 to 12.39/8.00 mmHg.

Selection context: plain CNN has the slightly better mean validation score; SE is our retained V2 backbone, not the winner of every ranking.

[7] Three-seed mean absolute errors on the same corrected splits. CBAM means Convolutional Block Attention Module. Temporal and additive gates are alternative weighting modules. Bold indicates numerical rank.

Group models: two signals versus one signal

Main-test pooled metric	ECG+PPG CNN+SE	PPG-only CNN+SE
MAE	12.39 / 8.00	**12.32** / **7.85**
RMSE	15.89 / 10.19	**15.73** / **9.88**
Bias	**-1.76** / **-0.89**	-2.05 / -1.64
Error SD	15.78 / 10.15	**15.57** / **9.67**
≤ 5 (%)	**26.03** / **38.83**	25.52 / 38.39
≤ 10 (%)	**49.22** / 68.91	48.67 / **69.38**
≤ 15 (%)	67.60 / 86.68	**67.82** / **87.85**

Values are SBP / DBP. Bold marks the best endpoint in this table. Lower errors, Bias closer to zero, and higher coverage are better.

PPG-only has slightly lower main-test MAE. Dual input has Bias closer to zero and the better validation score within the SE family.

On the separate AAMI-designated stress subset, macro MAE is **19.99/12.62** for dual versus **20.80/13.19** for PPG-only.

[7] Three-seed averages. Neither input mode wins every metric or dataset. We retain both.

Matched comparison: all 30 people

Equal-person MAE, then average of three seeds. Values are SBP / DBP, mmHg.

Method	PPG only	ECG+PPG
Population (no calibration)	14.66 / 10.60	14.80 / 10.35
Personal mean	12.24 / 6.02	12.24 / 6.02
Bias correction	8.51 / 5.38	9.09 / 5.70
Fixed PairSE	7.75 / 4.80	8.78 / 5.05
Single-period selection	7.72 / 4.67	8.17 / 4.78
Head-only adaptation	7.59 / 4.54	8.39 / 4.89
Tail adaptation (adopted)	**6.89 / 4.29**	**7.14 / 4.31**
Scope selection (comparison)	7.15 / 4.23	7.23 / 4.52

Adopted: tail adaptation, with each person selecting their own settings.

Bold identifies the adopted method here; this is not a claim that it wins every endpoint. Scope selection has slightly lower PPG DBP MAE. All controls use matched calibration/test records.

How much does the adopted method improve?

Comparison	MAE reduction (mmHg)	Relative reduction
PPG only vs Personal mean	5.35 / 1.73	43.70 / 28.77%
PPG only vs Bias correction	1.62 / 1.09	19.00 / 20.34%
ECG+PPG vs Personal mean	5.11 / 1.71	41.71 / 28.37%
ECG+PPG vs Bias correction	1.96 / 1.39	21.54 / 24.37%

Both pressures improve in the overall 30-person mean in every source seed.
This is an overall result, not a guarantee for every participant.

Same participant weighting and same development/test records for each comparison; SBP / DBP.

PPG only: full metric panel

Metric	Personal mean	Bias correction	Tail adaptation
MAE	12.24 / 6.02	8.51 / 5.38	**6.89** / **4.29**
RMSE	14.32 / 7.11	10.12 / 6.17	**8.51** / **5.16**
Signed Bias	-6.07 / -2.80	-2.81 / -1.48	**-0.70** / **-0.74**
Sample error SD	9.13 / 4.69	7.12 / 3.89	**6.52** / **3.68**
$\leq$ 5 (%)	31.14 / 58.60	42.98 / 59.35	**47.14** / **65.91**
$\leq$ 10 (%)	56.26 / 81.76	70.29 / 86.43	**75.87** / **92.65**
$\leq$ 15 (%)	72.42 / 89.22	85.07 / 94.18	**90.50** / **98.91**

Values are SBP / DBP. Bold marks the best endpoint in this table. Lower errors, Bias closer to zero, and higher coverage are better.

Tail adaptation improves both errors and coverage relative to matched controls.

Thirty participants, three source seeds. Error SD is averaged within person; pooled results are reported separately in the IEEE report.

PPG only: does the result repeat across seeds?

Method	Seed 42	Seed 43	Seed 44
Personal mean	12.24 / 6.02	12.24 / 6.02	12.24 / 6.02
Bias correction	8.45 / 5.37	8.58 / 5.32	8.50 / 5.45
Head only	7.84 / 4.45	7.81 / 4.94	7.13 / 4.23
Tail adaptation	7.31 / 4.43	6.68 / 4.25	6.69 / 4.18
Scope selection	7.41 / 4.32	7.22 / 4.29	6.82 / 4.10

Tail adaptation beats both baselines on both pressures in every seed.
Tail MAE standard deviation across seeds: 0.36 / 0.13 mmHg.

Seeds change source training randomness, not participants, splits or budgets. Results are separate runs, not an ensemble.

ECG+PPG: full metric panel

Metric	Personal mean	Bias correction	Tail adaptation
MAE	12.24 / 6.02	9.09 / 5.70	**7.14** / **4.31**
RMSE	14.32 / 7.11	10.86 / 6.54	**8.86** / **5.22**
Signed Bias	-6.07 / -2.80	-2.54 / -1.26	**-1.57** / **-0.77**
Sample error SD	9.13 / 4.69	8.11 / 4.52	**7.18** / **4.00**
≤ 5 (%)	31.14 / 58.60	39.95 / 55.02	**47.01** / **66.76**
≤ 10 (%)	56.26 / 81.76	66.51 / 84.10	**74.28** / **92.23**
≤ 15 (%)	72.42 / 89.22	82.24 / 94.19	**88.56** / **98.42**

Values are SBP / DBP. Bold marks the best endpoint in this table. Lower errors, Bias closer to zero, and higher coverage are better.

Tail adaptation improves both errors and coverage relative to matched controls.

Thirty participants, three source seeds. Error SD is averaged within person; pooled results are reported separately in the IEEE report.

ECG+PPG: does the result repeat across seeds?

Method	Seed 42	Seed 43	Seed 44
Personal mean	12.24 / 6.02	12.24 / 6.02	12.24 / 6.02
Bias correction	9.01 / 5.65	8.88 / 5.55	9.39 / 5.90
Head only	7.94 / 4.71	8.95 / 4.85	8.28 / 5.12
Tail adaptation	7.16 / 4.44	7.23 / 4.26	7.01 / 4.23
Scope selection	7.38 / 4.58	7.38 / 4.44	6.94 / 4.53

Tail adaptation beats both baselines on both pressures in every seed.
Tail MAE standard deviation across seeds: 0.11 / 0.11 mmHg.

Seeds change source training randomness, not participants, splits or budgets. Results are separate runs, not an ensemble.

Tail adaptation versus choosing the scope

Adopted method: fix the trainable components to the change head + last CNN block + SE. Each person still selects learning rate and update count.

Scope-selection comparator: additionally let internal validation choose head-only or tail fitting. This was tested, but is not our retained policy.

Policy	PPG MAE	Dual MAE
Head-only selection	7.59 / 4.54	8.39 / 4.89
Tail selection (adopted)	6.89 / 4.29	7.14 / 4.31
Head/tail scope selection	7.15 / 4.23	7.23 / 4.52

Personal hyperparameter selection does not require changing the trainable scope.

SBP / DBP, mmHg. Both strategies allow a validation-selected zero-update source checkpoint. The latter is a fallback, not a third learned architecture.

Overall gains and individual outcomes answer different questions

Strict wins over both controls, both pressures	PPG	ECG+PPG
Seed 42 / 43 / 44	11 / 9 / 7	12 / 12 / 15
After averaging each person's MAE across seeds	10 / 30	13 / 30
The same person wins in all three seeds	3 / 30	10 / 30

- The overall mean improves in all seeds even though some individuals worsen.
- Removing the largest beneficiary as a sensitivity check still gives tail-minus-bias MAE of -0.85/-0.55 (PPG), -1.33/-1.00 (dual).
- The official results retain all 30 people; no test-based exclusion or retuning.

Previously examined public-data cohort: exploratory evidence, not new external validation. Each person's full results remain in the release.

Can the personal model use cuff measurements?

Yes at the input-format level. PairSE needs an earlier waveform window and its two known pressure values. Its prediction function does not accept an ABP waveform.

Required information	Current public-data experiment	Student application
Earlier signal window	Recorded ECG/PPG	Student's saved ECG/PPG
Earlier SBP and DBP	Labels derived from ABP	Paired cuff reading
Current signal window	Later recorded ECG/PPG	Student's later ECG/PPG
Current true pressure	Used only for scoring	Later cuff reading for evaluation

Changing the label source is possible. Its effect on accuracy requires evaluation.

[7] PairSE.forward(current, reference, reference_bp). Reference BP contains two scalar values. Student input compatibility does not establish student performance.

What a small cuff-calibration study must test

Whose data? One student's early pairs for calibration, separate later pairs for evaluation.

- Our adopted public-data experiment uses **all eligible early-80% reference windows**. A student with a few cuff readings supplies far fewer independent labels.
- Pair each cuff reading with the signal window specified by the collection protocol. A cuff estimate summarizes a measurement interval.
- Keep later cuff readings out of personal fitting. Use them to compare PairSE, personal mean and bias correction with the **same early readings**.
- Check settings for this smaller budget on development data before the final student evaluation.

The published results here concern dense public-data calibration.

Proposed transfer evaluation, not an experiment already completed. Interpolating between cuff readings would not create measured continuous ABP or additional independent pressure labels.

What we have, and what the results support

- **Population**: compact CNN+SE in PPG-only and ECG+PPG versions. Source training is unchanged.
- **One personal policy**: update the change head, last CNN block and SE. Each person selects settings separately for SBP and DBP using three internal temporal periods.
- **Overall personal MAE**: PPG 6.89/4.29; dual 7.14/4.31 mmHg. Both beat personal mean and bias correction in every seed.
- **Retain both inputs**: PPG has slightly lower mean MAE; dual is more stable across source seeds. No universal input winner is established.

20261015 release: code, endpoint weights, complete metrics and IEEE report

[7] The source model, each person's two endpoint checkpoints, and their own reference bank define the personal predictor. Earlier releases remain archived.

References: data and network building blocks

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CVPR paper; extended arXiv version.

The architecture diagrams in this presentation depict our implementation. They are not copied figures from these papers.

References: personal modeling and our implementation

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